Did Humans Invent Language?

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Abstract

No non-human animal has spontaneously acquired language with open-ended vocabulary and rule-governed syntax. This paper proposes that language was invented by early modern Homo sapiens—a cultural act by cognitively capable individuals. Three clusters of novel hypotheses follow from this invention thesis. First, the language invention cluster: the Evolutionary Barrier to Language Hypothesis posits that no species has evolved language by incremental natural selection alone; the Language Invention Hypothesis proposes that humans invented a Free-Form Protolanguage (FFP) of symbolic words and flexible word combinations without structured syntax—sufficient to convey whole thoughts; and the Two-Stage Language Origin Hypothesis holds that this culturally invented FFP preceded and drove the biological evolution of recursive grammar. Second, the information-theoretic consequences of invention: the Natural Language Codec Hypothesis proposes that vocabulary and grammar function as compression algorithms, producing an effective data rate (EDR) of spoken language far exceeding the acoustic channel rate of ~39 bits/s (Coupé et al., 2019). Third, the biological consequences of invention: the Rapid Language Diffusion Hypothesis posits that language provided decisive competitive advantages, driving global dispersal of *H. sapiens*; the Language Lifespan Extension Hypothesis proposes that language drove extended human lifespan, generating two subsidiary hypotheses: the Grandparenting Hypothesis, in which knowledge transmission created selection for extended lifespan in both sexes, and the Fossil Menopause Hypothesis, in which menopause is a remnant of the pre-linguistic lifespan limit. Together, these hypotheses suggest that language was the most consequential invention in human history.

Introduction

Language is the signature trait of the human species—central to understanding our place in nature. Theories about its origin are necessarily speculative, and any account must be read with appropriate skepticism. The Société de Linguistique de Paris famously banned all papers on the subject in 1866, deeming the topic too speculative for productive scientific debate. Today, advances in anthropology, genetics, and cognitive science have made it scientifically viable. This paper attempts to mitigate speculation by grounding each hypothesis in established evidence, making explicit predictions where possible, and clearly distinguishing theoretical proposals from empirical claims.

Most work on language origins treats language as a product of incremental biological evolution by natural selection. This view presents a difficulty: no other species has evolved anything resembling open-ended symbolic language, despite millions of years of opportunity. Language appears to have evolved only once in the entire history of animal life—suggesting that selection pressure alone was insufficient.

An alternative is that language was *invented* and that biological evolution followed rather than preceded it. Everett (2017) anticipated the idea that language was a creation of culture. Bickerton (1990) and Tomasello (2008) proposed protolanguage theories in which words preceded grammar. None of these accounts develops the invention hypothesis into a systematic, causally explicit model, nor examines the information-theoretic consequences of language invention, nor traces the biological changes it may have triggered downstream—including a dramatic extension of human lifespan itself. This paper proposes eight named hypotheses supported by over 60 peer-reviewed citations.

1 Animal Communication

No non-human animal has acquired language with open-ended vocabulary and rule-governed syntax (Hage, 2024). Vervet monkeys have innate, genetically determined alarm calls specific for leopards, eagles, and snakes (Seyfarth et al., 1980). Wild bonobos and chimpanzees combine pairs of innate calls (e.g., yelp-grunt) into compositional structures with modified meanings (Berthet et al., 2025; Girard-Buttoz et al., 2025), a form of primitive syntax without words.

Vocal production learning—the ability to imitate or invent sounds—is rare (Janik & Slater, 1997). Despite their cognitive sophistication, wild chimpanzees do not mimic sounds and cannot invent new symbolic vocalizations. Attempts to teach chimpanzees spoken language (Hayes & Hayes, 1951) and sign language produced limited results—apes learned signs but showed no evidence of productive syntax or spontaneous symbol creation (Gardner & Gardner, 1969; Terrace et al., 1979). With intensive training, bonobo Kanzi used ~400 lexigrams and reportedly understood ~3,000 spoken English words (Savage-Rumbaugh & Lewin, 1994), yet showed no evidence of recursive syntax or open-ended word coinage. Critically, Kanzi could not create new symbols to represent concepts for which he lacked lexigrams.

No wild animal has been documented spontaneously using a learned symbolic system with syntax. Wild animals can sometimes learn elements of such systems when humans teach them—but none has invented one. This distinction between taught and invented symbolic communication lies at the heart of the invention hypothesis developed in Section 5.

Given its overwhelming competitive advantages, why has only one species achieved language? Evolutionary barriers range from low to insuperable. Eyes independently evolved more than 40 times. Powered flight evolved only four times. Non-avian dinosaurs did not evolve language in over 160 million years. In over 66 million years since the K–Pg extinction event, no mammal lineage evolved language until *H. sapiens*. The evolution of language by incremental natural selection alone may have been impossible. The Evolutionary Barrier to Language Hypothesis is summarized in Section 13.

2 Is Language Needed for Thought?

Some linguists argue that the original function of language was for reasoning rather than communication. Chomsky proposed that language evolved as a "tool for thought" (Berwick & Chomsky, 2016). He maintained that "externalization" (vocal speech) evolved later and was secondary to internal cognition. Other linguists argue that language is primarily a tool for external communication (Fedorenko et al., 2024).

High-level cognition may have preceded language. Many forms of conscious thought do not require language. *H. habilis* made stone tools approximately 2.6 million years ago (Klein, 2000). The large brains of *H. heidelbergensis*, *H. neanderthalensis*, and early *H. sapiens* may have evolved before language to process and remember environmental knowledge, and to facilitate tool-making and organized hunting with weapons (Stout & Chaminade, 2012). Some argue that the deep structures of human cognition are not linguistic (Fedorenko et al., 2024) and that language primarily functions to provide a posteriori explanations of concepts to ourselves and to others. Neanderthals may have been capable of brilliant insights, but without language those insights died with them. Language serves to organize, communicate, and preserve ideas across generations, rather than leaving them locked inside the heads of isolated individuals. Unlike episodic social learning—which resets with each generation—language enabled cumulative culture: a ratchet effect in which innovations are preserved, transmitted, and built upon across generations (Tomasello, 2008). This may explain the accelerating pace of technological change documented in the Upper Paleolithic record.

Vocal communication may be a primary, not a secondary, function of language. Through recursive syntax, we can express endless, complex, abstract thoughts (Jackendoff, 1983). The orderly, near-effortless turn-taking of human conversation (Sacks et al., 1974) enables the collaboration that underlies human culture.

3 Early Hominin Communication

There is ongoing debate about when the first elements of language appeared. Some argue that *Homo erectus* may have had early pre-language around 2–1.5 million years ago (Deacon, 1997). A more recent possible ancestor, *Homo heidelbergensis*, may have had pre-language about 400,000 years ago. *H. heidelbergensis* was also likely ancestral to *Homo neanderthalensis*. The most common estimates for the emergence of a protolanguage range from about 100,000 to 80,000 years ago. The evidence for early protolanguage includes:

- Brain size of *H. heidelbergensis* and Neanderthals similar to modern humans;
- Hyoid bone shape in *H. heidelbergensis* and Neanderthals similar to humans;
- Neanderthals had a FOXP2 gene (required for fluent speech) identical to humans.

None of the above are definitive for pre-language. Brain organization and shape may be more important than volume. Vocal tract anatomy does not determine speech capability—functional languages could have restricted sound inventories (e.g., limited phonemes). The FOXP2 gene is not unique to hominins; related versions appear in chimpanzees, songbirds, bats, and crocodilians, and are highly conserved across vertebrates. FOXP2 is a necessary but not sufficient condition for language.

The appearance of functional human language may be marked in the archaeological record by a combination of characteristics:

- prolonged infancy and childhood
- high rate of technological innovation
- symbolic complexity and behavioral modernity
- long-distance trade and exchange networks
- rapid population growth
- outward migration
- rapid extinction of other *Homo* species

Recent evidence that Neanderthal infants experienced significantly faster somatic growth than modern human infants (Been et al., 2026) suggests that the slow early development characteristic of *H. sapiens* may be a specifically modern human adaptation—consistent with the extended critical period required for language acquisition (Lenneberg, 1967).

H. heidelbergensis and Neanderthals had very low rates of technological innovation, largely unchanged for about 200,000 years (Klein, 2000). Simple tool making (e.g., knapping an Acheulean hand ax) can be learned by observation and practice, without using language (Stout & Chaminade, 2012).

Symbolic cultural behavior in *H. sapiens* appeared at least 77,000 years ago in Africa with shell beads and engraved ochre (Henshilwood et al., 2002). This sudden acceleration in innovation may reflect the earliest effects of language on human culture.

Some linguists think that human language originated only once (Atkinson, 2011). Chomsky argued that the core computational property of language (Merge) arose from a single chance mutation in one individual within an ancestral *H. sapiens* population, subsequently spreading through the group (Berwick & Chomsky, 2016)—a disputed saltationist view of the evolution of syntax. There are no documented cases of fully formed human communities using partial or incomplete languages. Since all humans have equal linguistic capability, language likely originated in an ancestral African population before regional diffusion and cultural diversification (Lenneberg, 1967), consistent with the theory of a single origin of languages.

After about 45,000 years ago, modern humans in Europe progressed through a rapid succession of distinct tool traditions, increasing regionalization, miniaturization, composite tools, bone and antler working, needles, rope, nets, and eventually ceramics. The Upper Paleolithic Revolution (~50,000–10,000 years ago) saw an explosion of sophisticated tools, including stone blades, composite hafted tools, atlatl spear throwers, and cave paintings (Klein, 2000). This cultural flowering probably resulted from language.

The population *H. sapiens* expanded in Africa 80,000–60,000 years ago (Mellars, 2006; Henn et al., 2012) and out of Africa about 60,000–40,000 years ago, primarily via a coastal migration route (Macaulay et al., 2005), though multiple dispersal events are also supported by genomic evidence. All other *Homo* species went extinct 100,000–40,000 years ago, perhaps unable to compete with *H. sapiens*. Speaking humans became the most lethal apex predator in the history of the planet.

Taken together, the archeological evidence points to the advent of fully modern, syntactically complex language around 80,000–50,000 years ago. However, correlation does not prove causation.

4 Precursors for Language

Several biological and cognitive preconditions appear to have been necessary before language could be invented. Intentional gestural communication—pointing, iconic gestures, and facial expressions—may have established the joint attention and referential intent that symbolic words later exploited (Corballis, 2002; Tomasello, 2008). Vocal production learning—the ability to produce and modify novel sounds—was equally important (Janik & Slater, 1997). These capacities, while necessary, are insufficient alone; many species share them without inventing language.

Musicality—the biological capacity to perceive and respond to structured sound—may have been the most important precondition. Musicality arises from pre-existing auditory and cognitive capacities (Trainor, 2015), is partially shared with other vocal learning species (Patel, 2021), and involves neural circuits partially overlapping with those used for language (Peretz et al., 2015; Honing, 2026). Music-making—the culturally transmitted production of structured sound—is built on this foundation and appears unique and universal to humans (Honing, 2026; Fitch, 2006).

Invented music and invented words share a critical property: both require mapping arbitrary sounds to learned meanings, and both demand the fine voluntary vocal control that distinguishes *H. sapiens* from other primates. The recent finding demonstrates that Alston's singing mouse evolved expanded orofacial motor cortical projections to auditory regions, a neural circuit supporting complex structured vocalizations. This suggests that musicality and language share a common neural architecture that can evolve independently of general intelligence (Banerjee et al., 2026). Musicality may thus have been the biological scaffold on which the cultural invention of music, and ultimately language, was built.

These preconditions—joint attention, vocal production learning, and musicality—were necessary but not sufficient. Also needed was the creative capacity for metaphorical abstraction required to map arbitrary sounds to meanings. Cognitive modernity, present in *H. sapiens* by at least 100,000 years ago, made language possible. What remained was the cultural act of invention itself.

5 Was Language Invented?

The Language Invention Hypothesis posits that a small group of early modern humans may have had the cognitive capacity to invent (rather than genetically evolve) words with flexible and arbitrary definitions—a new form of coded, symbolic thought. A single creative individual could have coined many words, assigning abstract meanings to spoken sounds. New utilitarian words could have been copied and adopted by others—a local cultural innovation and the beginning of language as a shared skill. Humans still invent thousands of new words per year.

Structured syntax was not essential for an early, rudimentary protolanguage. Unstructured word groups can convey information and intentions effectively (e.g., *gazelle—valley—you—spear—kill*). In context, groups of words with flexible order often work well enough. This invented Free-Form Protolanguage (FFP) may have provided the first means to express complete ideas. The FFP was a learned, symbolic, referential communication system—a functional language without regular syntax.

The concept of a protolanguage was anticipated by earlier theorists. Bickerton (1990) argued that early humans gradually evolved a protolanguage with isolated words and simple phrases but no grammar. Tomasello (2008) similarly argued that words and intentional communication preceded grammatical structure, attributing this sequence to the gradual evolution of shared intentionality. The FFP hypothesis differs from both: it proposes that protolanguage was a *de novo* cultural invention by cognitively modern *H. sapiens*, not a relic of biological evolution in earlier hominins.

An FFP could have been widely adopted and elaborated across many generations, with an ever-expanding vocabulary and words concatenated in any order. Its spread within a social group would have been facilitated by the human capacity for joint attention and eye gaze following (Tomasello, 2008), enabling early language users to establish shared context for newly coined words. Even a rudimentary FFP would have provided survival advantages during the uniquely difficult and dangerous process of human childbirth (Trevathan, 2011). The FFP resembled “telegraphic speech”—used by children and adults in high-concentration tasks. Verb-noun compounds such as *scarecrow* have been identified as syntactic "living fossils" of simpler, earlier grammatical structures (Jackendoff, 2002; Progovac, 2015), consistent with the FFP as a functional protolanguage. Individuals deprived of linguistic input during the critical childhood period, including Genie (Curtiss et al., 1974) and documented deaf linguistic isolates, fail to develop recursive syntax but do acquire lexical and concatenative skills analogous to an FFP.

The FFP was functional, but limited. Complex expressions would have required ever-larger word groups, creating a combinatorial burden. Without syntax, an FFP lacked the expressive power and efficiency of language as spoken today.

Profound human inventions can trigger cascading evolutionary change. The invention of tools and cooking with fire may have remodeled human digestive anatomy and facial morphology (Wrangham, 1999). The use of fire may have driven the evolution of shorter, REM-concentrated sleep in *H. sapiens* (Samson & Nunn, 2015), freeing extended evening hours for language-mediated social bonding and knowledge transmission. The invention of the FFP constituted a niche construction event (Odling-Smee et al., 2003) of unprecedented cultural scope: by creating a self-modifying cultural environment, language drove the biological changes documented in subsequent sections.

Once the FFP was in use, strong selection pressure for regularized grammar may have followed. This second stage of language development may have required the incremental evolution of specialized brain structures for fluent speech (Progovac, 2015). Deacon (1997) argued that symbolic language use created selective pressure for the co-evolution of the prefrontal cortex—language pulling its own biology into existence. A biological disposition for language (Chomsky's "Language Faculty" or Universal Grammar) subsequently became embedded in the human genome.

The biological evolution of language was accompanied by further cascading changes. The descended larynx enabled richer phonemic production, at the high cost of increased choking risk. A prolonged juvenile period evolved to accommodate cultural knowledge and language acquisition. The uniquely flat, gracile face of *H. sapiens*—in contrast to the projecting mid-face and heavy brow ridges of Neanderthals and earlier hominins—may reflect selection for both the fine articulatory movements of speech and the paralinguistic facial expressions enabled by mobile eyebrows that enrich verbal communication (Godinho et al., 2018). Uttley et al. (2025) showed that modern humans carry a less active version of a non-coding SOX9 enhancer than Neanderthals, consistent with our more gracile mandibles and potentially contributing to the vocal tract reconfiguration for fluent speech. Refinement of precision hand grip morphology in *H. sapiens*—compared to the power-grip-optimized hands of Neanderthals—supported both fine tool use and the gestural communication that accompanied language (Greenfield, 1991).

The FFP may have been a necessary precondition for language to fully evolve. While large brains had evolved in *H. heidelbergensis* and *H. neanderthalensis*, only *H. sapiens* demonstrated the behavioral modernity that signals the cognitive threshold required for arbitrary symbolic invention, including artistic creation, long-distance exchange networks, and rapid technological innovation (Klein, 2000). Darwin (1871) proposed that sexual selection for eloquence—verbal ornamentation as a signal of cognitive fitness—may have been a significant driver of brain evolution for spoken language. Progovac (2026) proposed that sexual selection for quick-wittedness and humor played a causal role from the earliest stages of human language evolution.

The **Two-Stage Language Origin Hypothesis**—the cultural invention of a protolanguage followed by biological evolution of brain structures for recursive grammar—may explain why open-ended language with syntax did not evolve elsewhere in nature. The required sequence of cultural invention *preceding* biological selection may represent an evolutionary bottleneck that no other lineage has traversed. The two stages are illustrated in Fig. 1.

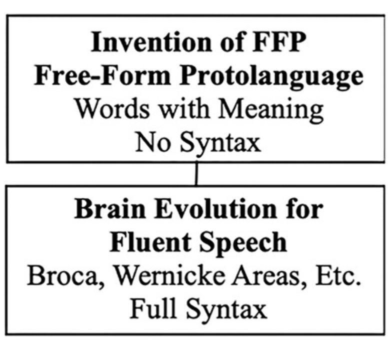

Fig. 1 Two-Stage Language Origin Hypothesis

The two stages of the Two-Stage Language Origin Hypothesis operated on very different timescales. Stage 1—the cultural invention of an FFP—could have been established within a few generations, around 100,000-80,000 years ago, consistent with the earliest symbolic artifacts. Stage 2—the biological evolution of brain structures for recursive syntax—would have required a substantially longer time. The African archaeological record provides an empirical constraint: the 20,000–30,000-year interval between the appearance of unambiguous symbolic artifacts (~80,000 years ago) and the Upper Paleolithic cultural explosion (~50,000 years ago) represents approximately 800–1,200 human generations, assuming a generation time of 25 years. Under strong selection pressure from FFP use, this may have been sufficient for fixation of multiple advantageous alleles affecting the neural circuitry of syntax.

A key implication of the Two-Stage Hypothesis is that Stage 1 required cognitive modernity but not modern brain shape—the FFP could be invented with the neural architecture already present in early *H. sapiens* by ~300,000 years ago. The globularization of the cerebellum and parietal lobe—structures central to language processing and articulation—was completed gradually between ~100,000 and ~35,000 years ago (Neubauer et al., 2018), coinciding with the proposed timing of Stage 2. Multiple selection pressures may have contributed, including dietary change and expanded social complexity. Language is also a plausible explanation: the specific globularized structures are those most directly implicated in language processing. A single new selection pressure acting on both simultaneously is simpler than invoking multiple independent causes.

This incremental brain evolution is consistent with the gradual emergence of syntactic complexity: Progovac (2015) argues that present-day languages retain fossil remnants of earlier, non-hierarchical proto-syntactic stages, consistent with syntax having evolved in layers rather than in a single step. The gene-culture feedback loop proposed here—FFP use creating selection pressure, which drove brain evolution, which enabled richer syntax, which created further selection pressure—would have been self-amplifying, potentially compressing the Stage 2 timeframe.

Child language acquisition follows the same sequence proposed here for the evolution of language: lexical-semantic productivity—words combined to convey whole thoughts without grammatical structure—consistently precedes the acquisition of recursive syntax (Tomasello, 2008). Every child passes through an FFP-like telegraphic stage before acquiring full grammar, suggesting that the lexical foundation proposed in Stage 1 is both developmentally and evolutionarily prior to the syntactic structures of Stage 2.

6 Natural Language Codec Hypothesis

The invention of words for the FFP required the cognitive capacity for metaphorical abstraction (Lakoff & Johnson, 1980). De novo word coinage maps a concrete reality onto an arbitrary sound, compressing all associated meaning into a single symbolic token. Invented language is thus not merely a labelling system but a novel Natural Language Codec (NLC) communication system.

Codecs increase the information transmitted per unit time. Computers use codec algorithms to compress and decompress data (MP3, for example, compresses music for streaming and playback). Encoding and decoding are separate operations, which may occur at different locations or times. Codecs can be lossless (no data lost) or lossy (some data lost to achieve greater compression), and are characterized by a compression ratio (CR) (Watkinson, 2002). Biological codec systems are also found in nature: the immune system encodes information about past pathogens in memory B cells, later decoded to manufacture neutralizing antibodies; DNA stores genetic information selectively decoded via transcription and translation to build and maintain a living organism.

Human speech and perception operate as complementary codec systems—the speaker compresses meaning into sound, the listener decompresses sound into meaning, using shared vocabulary and grammar as the codec algorithms. NLCs overcome the inherent information-rate limits of the speech channel: Shannon (1948) established the information-theoretic framework later used to estimate the speech information rate (IR). Coupé et al. (2019) used syllabic information density and speech rate to measure a consistent IR of ~39 bits/s across 17 languages from 9 language families—a plausible universal limit for spoken language. Words carry symbolic meanings that transcend this limit: prior analyses of speech information rate have not accounted for lexical meaning (Crocker et al., 2016) or semantic content. NLC algorithms therefore define an effective data rate (EDR)—a semantic expansion metric for information density (not a Shannon information rate)—distinct from the acoustic information rate. IR and EDR are both specified in bits/s but are not equivalent. Semantic expansion estimates should correlate with comprehension latency, word frequency, concreteness, and contextual predictability.

Both individual words and word groups constituted novel codecs. An FFP featured groups of words without fixed grammatical rules; this combinatorial productivity amplified semantic content, allowing early language users to express intentions and coordinate actions far beyond the capacity of individual signals. Unlike biological codecs, which evolved over millions of years, Natural Language Codecs were invented and refined culturally, at a pace far exceeding biological evolution—a distinction that may explain the sudden appearance of behaviorally modern humans in the archaeological record (Klein, 2000).

The two principal NLCs are analyzed below: the Lexical Codec, operating at the level of individual words, and the Syntactic Codec, operating at the level of structured sentences. Example EDR calculations are provided for each. NLC theory is not a replacement for other formal theories of language.

7 Lexical Codec

The initial inventive step toward language was the lexical codec. What seems obvious today, that words have meaning, was completely novel at the dawn of language. An arbitrary utterance was assigned a symbolic meaning, such as a name for a person, an object, or an action—the beginnings of a lexicon. For this invention to succeed, a newly coined word had to be learned and understood by listeners, requiring vocal learning ability.

Humans use vocabulary to partially compensate for the low IR of the speech channel (Jackendoff, 2010). Memorized vocabulary contains compressed information about the meaning of words. The single-syllable word *horse* refers to a large four-legged animal; the superficially similar word *house* refers to a dwelling with a roof, windows, and a door. These seemingly simple observations form the foundation of the Lexical Codec. Vocabulary is a flexible NLC for compressing and decompressing word meaning, allowing near-effortless vocalization or recognition of approximately 20,000 English working vocabulary words.

Coupé et al. (2019) used the syllable as a unit for measuring the IR of various languages. Following that precedent, the syllable is also used as the information-encoding unit for CR and

EDR calculations. Counting syllables allows estimation of semantic content and calculation of the EDR of spoken language.

The word *horse* has one syllable. A short dictionary definition requires 36 syllables: "a large plant-eating domesticated mammal with solid hooves and a flowing mane and tail, used for riding, racing, and to carry and pull loads"—a syllabic compression ratio of 36:1. Speaking time confirms this: *horse* takes approximately 0.3 seconds; its definition approximately 11 seconds (CR = 11 ÷ 0.3 ≈ 36).

The CR can be variable: for a person with real-world experience of horses—their temperament, mass, movement, and smell—the word conjures a far richer representation than a dictionary definition captures. If the listener does not know what a horse is, or a sparse definition is used—"a large hoofed mammal" (5 syllables)—then the CR and information communicated are both reduced. Semantic richness is thus approximated by definition length, and the EDR calculation below represents a conservative estimate.

The CR measures how many syllables of meaning are compressed into a single syllable. Multiplying the speech channel IR by the CR gives the effective data rate:

EDR = IR × CR = 39 bits/s × 36 = 1,404 bits/s

The full meaning of *horse* is evoked in the listener's mind in a fraction of a second. Word identification occurs with a neural latency of approximately 114 milliseconds from phoneme onset (Brodbeck et al., 2018). The same advantage applies to *house*, or any other known word in the lexicon.

The 36-syllable definition of *horse* itself uses words, each expandable with its own definition—a recursive inflation that is unbounded, suggesting the true EDR substantially exceeds 1,404 bits/s, which represents a conservative lower bound. Memorized lexical entries also encode likely parts of speech, further increasing EDR. This analysis is nonetheless restricted to conservative first-order calculations.

8 Syntactic Codec

The Syntactic Codec is an NLC that compresses meaning of both FFP word groups and structured sentences. It is not a replacement for generative grammar or other theories of syntax. As with the Lexical Codec, the semantic gain of the Syntactic Codec can be expressed as a compression ratio, quantifying the EDR achieved.

The protolanguage FFP word group introduced earlier, the phrase *gazelle—valley—you—spear—kill*, has 7 syllables. The dictionary definition of gazelle, "a small, slender antelope that has curved horns and a fawn-coloured coat with white underparts," has 21 syllables. The definition of a valley, "a low land between hills," has 6 syllables. The definition of you, "the person that the speaker is addressing," has 11 syllables. The definition of a spear, "a long shaft with a pointed tip," has 8 syllables. The definition of kill, "cause the death of an animal," has 8 syllables. The combined syllable count for all five definitions is 54. The CR is therefore 54:7 or 7.7:1. These values are definition-dependent and intended only as order-of-magnitude illustrations.

The EDR is calculated by multiplying the universal IR by the codec CR:

FFP EDR = IR × CR = 39 bits/s × 7.7 = 301 bits/s

Just as the Lexical Codec requires learning thousands of words, the Syntactic Codec requires grammatical competence and cultural knowledge of a specific language. Consider the sentence, "Einstein formulated relativity theory"—four words with a total of 13 syllables. When each word is unpacked using partial Wikipedia and dictionary definitions, Einstein is described using 277 words (506 syllables). Formulated uses 10 syllables. Relativity theory uses 59 words (135 syllables). The combined syllable count for all three definitions is 651. The CR is therefore 651:13 or 50:1.

As before, the EDR is calculated by multiplying the universal IR by the codec CR:

EDR = IR × CR = 39 bits/s × 50 = 1,950 bits/s

This is more than 50 times the universal IR for human speech of ~39 bits/s (Coupé et al., 2019). The CR and EDR of the Syntactic Codec are instantaneous measures; both will vary from moment to moment, with language, spoken word rate, sentence complexity, and other factors.

Another example sentence concerns an abstract idea: “Meaning is a property of consciousness” (Brian Cox, UK physicist and science communicator). Here there is no mental image of a physical object—the sentence elicits thoughts about how perception depends on awareness. Consciousness is an elusive concept; since the speaker is communicating abstract philosophical ideas, the syntactic CR may be indeterminate or immeasurably large.

Fig. 2 shows a conceptual flowchart of a general Syntactic Encoder Algorithm for an NLC. A speaker first conceives the thoughts and emotions to be expressed. Vocabulary words are then selected from a memorized lexicon, aided by context and the part-of-speech tag (e.g., noun, verb) associated with each word. The selected words are encoded and compressed to generate a syntactically correct sentence—a process that may embody Chomsky's Merge or some other mechanism of generative syntax. Finally, the sentence is spoken with the desired emotional tone, employing the approximately 100 coordinated muscles involved in fluent speech.

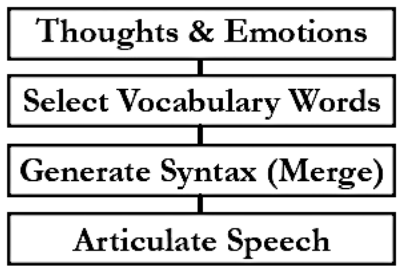

Fig. 2 Syntactic Encoder Algorithm

Chomsky's (1995) Minimalist Program posits that Merge is a fundamental syntactic operation combining two syntactic elements into a new hierarchical structure—this can be interpreted as an NLC compression algorithm. However, Merge describes only half of a Syntactic Codec; it specifies an encoder but not a decoder. The goal of syntax decoding is for the listener to understand what has been spoken. Although Merge has been influential, alternative models of generative syntax exist (Friederici, 2011).

Fig. 3 shows a conceptual flowchart of a general Syntax Decoder Algorithm for an NLC. A meaningful communication begins with spoken words. The listener matches each word to a memorized vocabulary entry, along with its part-of-speech tag. Drawing on a shared language, a predicted syntactic structure (Nakamura et al., 2026), and context, the listener can then parse the meaning of the sentence.

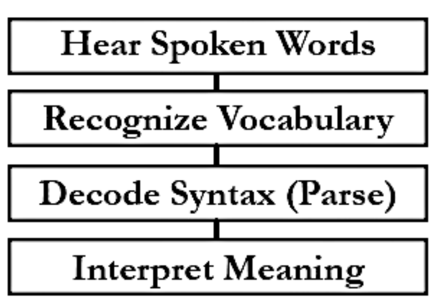

Fig. 3 Syntax Decoder Algorithm

Parsing syntax is remarkably rapid, even for high-EDR sentences. When English sentences follow proper subject-verb-object word order, listener brain activity can be detected within 130 milliseconds—less than an eye-blink (Fallon & Pylkkänen, 2024). This fast processing is enabled in part by Wernicke's area (left temporal language networks), which extracts meaning from heard speech within 50–150 ms of perception. Typical processing of 100–400 milliseconds per word is what makes fluent conversation possible.

9 Matching Codec Encoders and Decoders

Computer codecs usually have an identifying suffix—such as MP3 for audio, JPG for images—that ensures the correct codec algorithm is used to compress, decode, and display the data. If data transmission errors occur, error correction routines can often recover corrupted or incomplete data.

NLCs are asymmetrical (for encoding and decoding) and specific to a particular language. Like computer codecs, the NLC encoder and decoder must match: speaker and listener must share the same language. Native speakers can also predict sentence structure (Nakamura et al., 2026), aiding rapid comprehension. Most English speakers tolerate variation and error: when a speaker uses an accent, mispronounces a word, or makes a grammatical mistake, meaning is usually still conveyed—the natural-language equivalent of error correction—though misunderstandings do occur.

For conversational English, a speaker can adjust vocabulary and grammar to suit the listener—a teacher simplifying language for a child, or a professional using specialized jargon with a peer (e.g., physicians discussing a diagnosis). Beyond shared language, successful verbal communication often depends on shared cultural knowledge.

Native speakers of English can readily gauge another speaker's verbal competence. The voice conveys tone, mood, intention, and identity; a trained ear can further detect geographic origin, education, social class, and other social and cultural background markers.

10 Rapid Language Diffusion Hypothesis

NLCs may have driven rapid spread of language-speaking people. The first group of early modern humans to achieve language likely discovered its overwhelming competitive advantages, unprecedented in the natural history of Earth. No other species can long withstand human ecological dominance (Shea, 2023).

When language first appeared between 100,000 and 70,000 years ago, the various African *Homo* populations were sparse and scattered (Hollfelder et al., 2021). However, there was a population explosion in Africa between 80,000 to 60,000 years ago (Mellars, 2006) when humans spread to diverse habitats (Hallett et al., 2025), from forests to arid deserts, perhaps due to the adaptive advantages provided by language.

Independent genomic evidence supports this rapid dispersal model: Y-chromosome analysis places a cluster of major non-African founder haplogroups in a narrow window of 47,000–52,000 years ago, consistent with rapid initial colonization of Eurasia and Oceania following an out-of-Africa bottleneck (Karmin et al., 2015)—within the window proposed here for the advent of fully modern language.

An out-of-Africa dispersal beginning around 60,000 years ago led to the permanent establishment of *H. sapiens* populations in Europe, followed by the disappearance of Neanderthals. Burke et al. (2026) provide independent spatial modelling evidence that *H. sapiens* social networks in Europe were substantially more connected than those of Neanderthals during the critical period of 60,000–35,000 years ago. Language is a plausible explanation for this connectivity advantage: the exchange of information about distant resources and the maintenance of alliances required a symbolic communication system. Language also provided the ability to plan, remember, and communicate about non-present events (Suddendorf & Corballis, 2007). Language may have enabled *H. sapiens* to outcompete Neanderthals.

After humans acquired language, verbal humans may have displaced all non-speaking African *Homo* groups (including non-verbal early modern humans). Speaking *H. sapiens* outcompeted, extirpated, or merged with all other relict *Homo* species, including Neanderthals. This serial founder effect (Henn et al., 2012) may explain why there is only a single human species today.

The sudden advent of an FFP in Africa may have occurred within a single extended family or band. Even an early FFP, without recursive syntax, may have provided significant competitive advantages. If so, it is possible to estimate the population growth of language speakers in Africa 80,000–60,000 years ago. The exponential population growth formula is:

$$x(t) = x_0 \cdot \left(1 + \frac{r}{100}\right)^t$$

where t is the number of years; x_0 is the initial population, and r is the percent growth rate.

As a hypothetical thought experiment, a very small founding population (~10 people), with an exponential growth rate of 1% (doubling time of less than 70 years, consistent with favorable conditions among modern hunter-gatherers), would have produced a total population of about 3,500 fluent language speakers within 600 years. This compares with genetic estimates that the effective ancestral African population numbered only a few thousand individuals around this period (Henn et al., 2012), consistent with a very small founding language-speaking group, before humans migrated out of Africa. The African population expansion may have had a lower growth rate and taken longer, but was still relatively rapid.

Proposed Language Timeline:

- ~300,000 years ago—modern brain *size* established; cognitive modernity present
- ~100,000-80,000 years ago—earliest symbolic artifacts; FFP invention proposed; brain globularization begins
- ~80,000–50,000 years ago—syntactically complex language emerging; Upper Paleolithic transition
- ~35,000 years ago—brain globularization complete; fully modern brain shape

Taken together, the evidence points to the advent of fully modern, syntactically complex language around 80,000–50,000 years ago. However, globularization was not exclusively language-specific and correlation does not prove causation.

11 Did Language Change the Human Lifespan?

Language's consequences may have extended beyond competitive advantage to the most fundamental biological parameter of all—lifespan itself. The invention of language not only drove a cultural flowering (Klein, 2000) but may also have extended human longevity. Wild chimpanzees have a mean adult lifespan of approximately 30 years (Hill et al., 2001), and Neanderthals and early modern humans appear to have fared similarly, with few individuals surviving beyond 40 years (Trinkaus, 2011; Caspari & Lee, 2004)—suggesting that short adult lifespans were the ancestral condition across hominins and our closest living relatives alike. In contrast, during the Upper Paleolithic transition about 50,000–30,000 years ago, the proportion of *H. sapiens* individuals surviving to older adulthood increased roughly fivefold (Caspari &

Lee, 2004), a shift with no parallel among Neanderthals. While the dates are debated, there is consensus about the reality and suddenness of this transition.

A species-typical lifespan reflects a heritable life-history trade-off between reproductive rate and body size. Species with high reproductive rates have short lives; long-lived species have low reproductive rates. Modern humans are anomalous outliers on this fast-slow continuum, with exceptional longevity relative to body size. The usual explanation is the Grandmother Hypothesis (Hawkes et al., 1998; Aimé et al., 2017), which postulates that post-reproductive females enhanced inclusive fitness by provisioning grandchildren, a self-reinforcing feedback loop promoting longer female lifespans.

The Language Lifespan Extension Hypothesis proposes a more fundamental driver. By enabling the accumulation and transmission of intergenerational survival knowledge—medicinal plants, safe foods, predator behavior, territorial resources, cooperative hunting strategies, and midwifery—language reduced extrinsic mortality in both sexes. Life history theory predicts that reduced extrinsic mortality creates selection pressure for extended somatic lifespan (Kirkwood, 1977): when the background rate of death falls, investment in bodily maintenance becomes evolutionarily worthwhile. The hypothesis is testable: the timing of lifespan extension should correlate with archaeological proxies for cumulative culture, long-distance exchange, and complex toolkits.

12 Grandparenting, Menopause, and the Fossil Lifespan

The Language Lifespan Extension Hypothesis generates two subsidiary hypotheses about the biological consequences of extended lifespan.

The Grandparenting Hypothesis generalizes the Grandmother Hypothesis (Hawkes et al., 1998). Language reduced extrinsic mortality, creating selection for extended lifespan in men and women equally. The fitness benefits of grandmothering (Hawkes et al., 1998) are real but likely represent a secondary reinforcing feedback on an extended lifespan. Post-reproductive individuals of both sexes—grandmothers and grandfathers—became repositories of linguistically transmitted survival knowledge. Grandparenting, broadly construed, became a powerful fitness-enhancing strategy because language made accumulated knowledge transmissible. The cooperative breeding and childcare system proposed by Hrdy (2009) exemplifies the extended social support, including grandparents of both sexes, that language facilitated.

The Grandparenting Hypothesis gains additional biological support from evidence that paternal caregiving in men is hormonally mediated. Storey et al. (2000) showed that expectant and new fathers experience hormonal shifts—rising prolactin, falling testosterone—that parallel those of mothers and prime males for infant care. Hrdy (2024) argues that the capacity for primary caregiving is a latent trait in all human brains, an "alloparental substrate" awaiting activation by proximity to infants. These findings suggest that grandfathers who remained closely involved with grandchildren in ancestral societies would have been biologically primed for caregiving investment.

In most mammal species, reproductive lifespan equals somatic lifespan—females remain fertile until death. Wild great apes and Neanderthals had mean adult lifespans of approximately 30–40 years—most females dying before or around age 50, the typical age of menopause in modern humans. The extended post-reproductive lifespan characteristic of *H. sapiens* is without parallel among our closest relatives living under natural conditions. Human female reproductive years are constrained by a finite ovarian reserve established at birth and progressively depleted throughout life. The Fossil Menopause Hypothesis proposes that menopause is a remnant of the previous lifespan limit: the point at which ovarian reserves were exhausted in a shorter-lived ancestor. Language-driven lifespan extension pushed beyond this ancestral limit, exposing menopause as a constraint that had previously been invisible. Menopause is the point at which ovarian reserves run out in a body that now lives twice as long as its ancestors.

13 Discussion and Original Contributions

This paper proposes eight hypotheses about language origins and the biological consequences of language, spanning evolutionary theory, information science, and human biology. The eight hypotheses, each stated below in relation to prior work, are:

- Evolutionary Barrier to Language Hypothesis
- Language Invention Hypothesis
- Two-Stage Language Origin Hypothesis
- Natural Language Codec Hypothesis
- Rapid Language Diffusion Hypothesis
- Language Lifespan Extension Hypothesis
- Grandparenting Hypothesis
- Fossil Menopause Hypothesis

Evolutionary Barrier to Language Hypothesis. In the history of life on Earth, no species—including *H. sapiens*—has evolved language solely by incremental natural selection. Language confers decisive fitness advantages: it enables social cooperation and cumulative knowledge transmission. If language were simply a product of natural selection acting on large, social brains, it should have evolved repeatedly. Yet in the entire history of animal evolution, language has appeared only once, within the last 100,000 years. This paper proposes that language faces an evolutionary barrier: incremental natural selection cannot produce arbitrary symbolic reference de novo, because no intermediate stage between animal signaling and symbolic language confers sufficient fitness benefit to be selected for. *H. sapiens* overcame this barrier not through biological evolution alone but through cultural invention—the deliberate creation of a Free-Form Protolanguage that bootstrapped the biological evolution of recursive grammar.

Language Invention Hypothesis. Humans overcame the evolutionary barrier by inventing a Free-Form Protolanguage (FFP): arbitrary symbolic words and flexible word combinations without grammatical structure, sufficient to convey whole thoughts and coordinate action. A single cognitively modern individual could have coined the first words—mapping arbitrary sounds to specific meanings—and others could learn and adopt it. This bootstrapping process required no biological change: the FFP ran on cognitive hardware already present in early *H. sapiens*. Prior protolanguage theories (Bickerton, 1990; Tomasello, 2008) attributed words-before-grammar to gradual biological evolution; this paper proposes deliberate cultural invention.

Two-Stage Language Origin Hypothesis. Everett (2017) argued that language was a cultural creation, and protolanguage theories placed words before grammar—but neither developed these insights into a causally explicit two-stage model. Stage 1 is the de novo cultural invention of a Free-Form Protolanguage (FFP) by cognitively modern *H. sapiens*: a combinatorial system demonstrating that semantic productivity does not require grammatical structure. Stage 2 is the subsequent biological evolution of brain structures for recursive grammar, driven by selection pressure from FFP use. This causally ordered sequence—cultural invention preceding biological evolution—may explain why open-ended language has not evolved elsewhere in nature: the conjunction of cognitive modernity, cooperative social structure, and deliberate symbolic invention represents an evolutionary bottleneck that no other lineage has traversed.

Natural Language Codec Hypothesis. Language was not merely vocalization—it was a novel codec communication system for compressed abstract meaning. Prior information-theoretic analyses of speech measure acoustic channel capacity at ~39 bits/s (Coupé et al., 2019), without accounting for the semantic content unlocked by shared vocabulary and grammar (Crocker et al., 2016). This paper introduces Natural Language Codecs (NLCs) to quantify that semantic compression. The Lexical Codec shows that individual words compress semantic content at ratios far exceeding the raw channel rate; the Syntactic Codec extends this to structured sentences, yielding EDRs orders of magnitude higher still. Because NLCs were culturally invented rather than biologically evolved, they developed far faster than natural selection could produce—a property that may explain the suddenness of behaviorally modern humans in the archaeological record.

Rapid Language Diffusion Hypothesis. Invented language provided decisive competitive advantages—coordinated hunting, long-distance alliance networks, cumulative knowledge transmission, and rapid adaptation to new environments—that no non-linguistic competitor could match. The result was explosive: genetic evidence confirms a great demographic expansion beginning approximately 45,000–60,000 years ago (Henn et al., 2012), coinciding with the Upper Paleolithic cultural explosion and the advent of fully modern syntactic language. All other Homo species went extinct during this same window. While the demographic expansion is well-documented, its primary cause has not been identified—this paper proposes that language was the decisive driver.

Language Lifespan Extension Hypothesis. Wild chimpanzees and Neanderthals had mean adult lifespans of approximately 30–40 years—similar to pre-linguistic *H. sapiens*. During the Upper Paleolithic transition, human lifespans extended dramatically, with the proportion of individuals surviving to older adulthood increasing roughly fivefold (Caspari & Lee, 2004). This paper proposes that language was the primary driver: by enabling intergenerational transmission of survival knowledge, language reduced extrinsic mortality in both sexes. Life history theory predicts that reduced extrinsic mortality creates selection pressure for extended somatic lifespan (Kirkwood, 1977), providing the mechanistic link between language invention and longevity. The Grandmother Hypothesis (Hawkes et al., 1998) identifies real secondary feedback within this process—but applies only to post-reproductive females and mistakes a consequence of lifespan extension for its cause. The Language Lifespan Extension Hypothesis applies equally to both sexes and generates two subsidiary consequences: the Grandparenting Hypothesis and the Fossil Menopause Hypothesis.

Grandparenting Hypothesis. The Grandmother Hypothesis (Hawkes et al., 1998) identifies post-reproductive female provisioning as a driver of extended female lifespan—a real effect, but sex-specific and silent on the mechanism. This paper proposes a broader Grandparenting Hypothesis: language enabled the accumulation and transmission of survival knowledge across generations, creating selection for extended lifespan in both sexes equally. Post-reproductive individuals, grandmothers and grandfathers alike, became repositories of practical wisdom and expertise that only language could preserve and transmit. The

Grandmother Hypothesis is subsumed within this larger framework as secondary feedback on a lifespan extension whose primary driver was language.

Fossil Menopause Hypothesis. In most mammal species, reproductive lifespan equals somatic lifespan—females remain fertile until death. Menopause has typically been explained as an adaptation: post-reproductive grandmothers enhance inclusive fitness by provisioning grandchildren (Hawkes et al., 1998). This paper proposes the opposite—menopause is not an adaptation but a fossil, a remnant of the shorter pre-linguistic lifespan. The fixed ovarian reserve established in embryonic development evolved for a lifespan of ~35 years, which was the ancestral condition shared by wild great apes, hominins, and Neanderthals. Language drove somatic lifespan beyond this limit, exposing a reproductive constraint that had previously been invisible. Menopause is the point at which ovarian reserves run out in a body that now lives twice as long as its ancestors.

Summary Comments: These eight hypotheses are necessarily speculative—language origins cannot be directly observed, and correlation does not establish causation. The paper attempts to mitigate speculation by grounding each hypothesis in established evidence and making predictions where possible. The most directly testable predictions are: that brain globularization should correlate temporally with archaeological proxies for symbolic culture rather than with earlier anatomical changes; that lifespan extension in *H. sapiens* should co-occur with cumulative cultural complexity in the fossil record; and that no non-human animal trained on symbolic communication should demonstrate de novo word coinage without human scaffolding.

Future research directions include: quantitative testing of NLC compression ratios across languages and sentence types; genomic analysis of longevity-related alleles for signatures of selection in the Upper Paleolithic window; and comparative studies of post-reproductive lifespan in great ape populations living under natural versus protected conditions.

Together, these hypotheses suggest that language was not merely a communication tool but the most consequential invention in the history of life on Earth. The invention of language was a cultural act that bootstrapped its own biology, extended human lifespans, and enabled the global dominance of a single primate species within an evolutionary eyeblink.

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